Calculating Non-resonant Inelastic X-ray Spectra from First Principles

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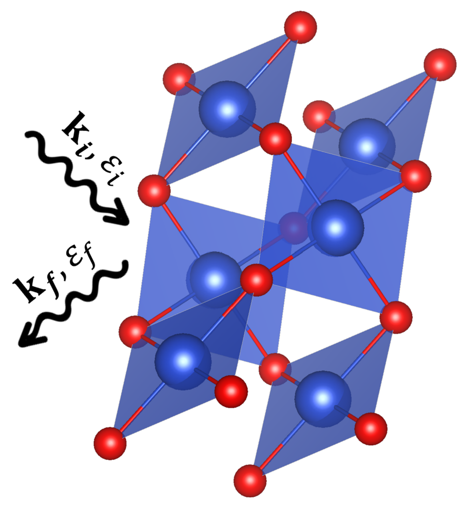
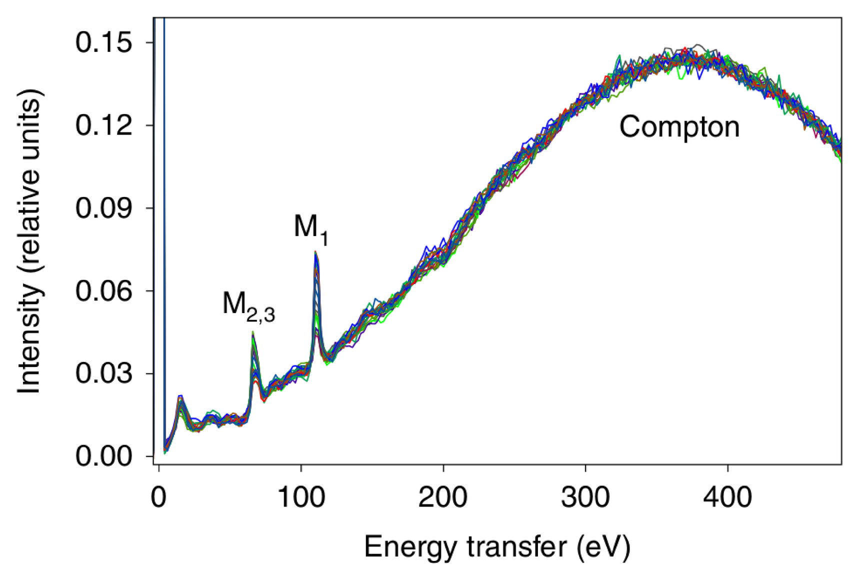
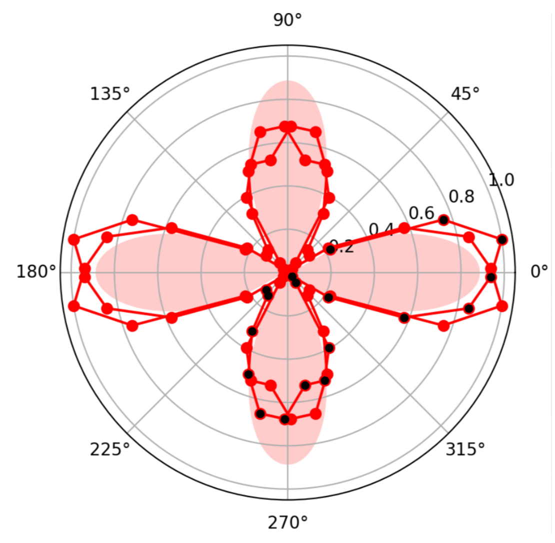
Motivation

Non-resonant inelastic x-ray scattering (NRIXS) gives access to the orbital shape of the hole-part of the material's ground state [1, 2]. Here, we aim to compute it from *ab initio* methods for CuO.







The M_1 edge can be used to probe quadrupole $3s \to 3d$ electronic transitions. The angular dependence is then dictated by the final state and we can therefore get an image of the Cu^{2+} hole.

Hubbard model

Copper oxide being a **strongly correlated material**, an appropriate first approximation model is the **one orbital Hubbard model** :

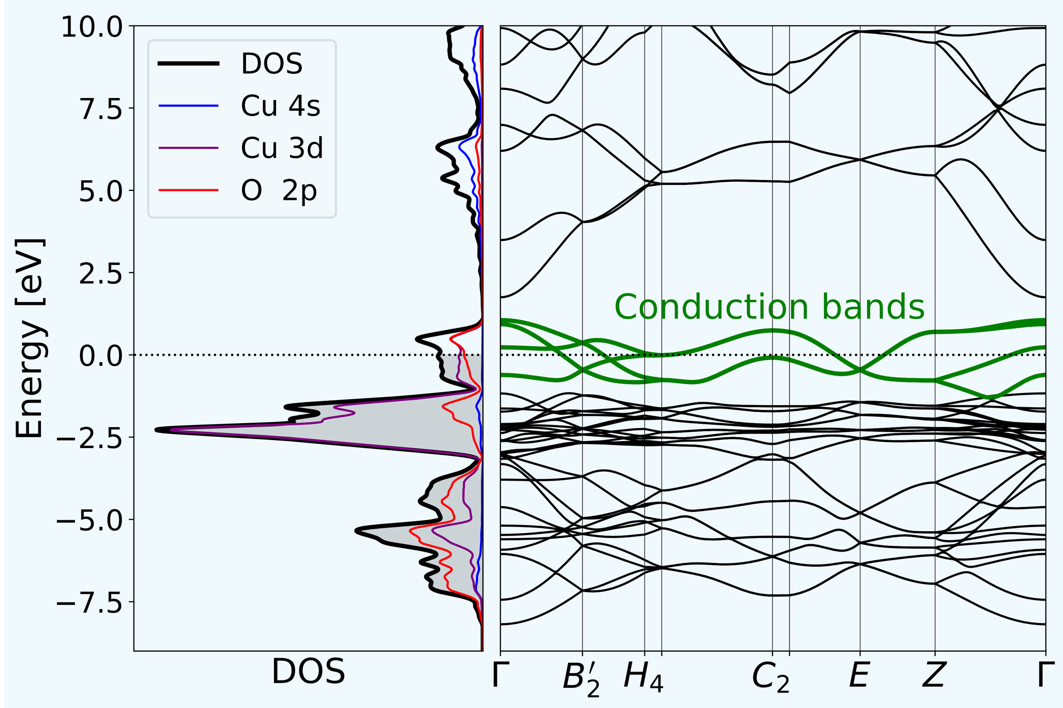
$$\hat{H} = -\sum_{ij\sigma} t_{ij} c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

where t_{ij} is the hopping amplitude between sites i and j, which tends to delocalize the electrons, and U is the on-site Coulomb repulsion, which favors their localization.

But which orbital should we choose? How can we compute t_{ij} and U for CuO? And how can we solve this model numerically ?

Density Functional Theory (DFT)

DFT is a mean-fied, ground-state theory. It allows to compute the Bloch states (BS), the bands (right) and the density of states (left) of CuO with projected Cu-3d/4s and O-2p.



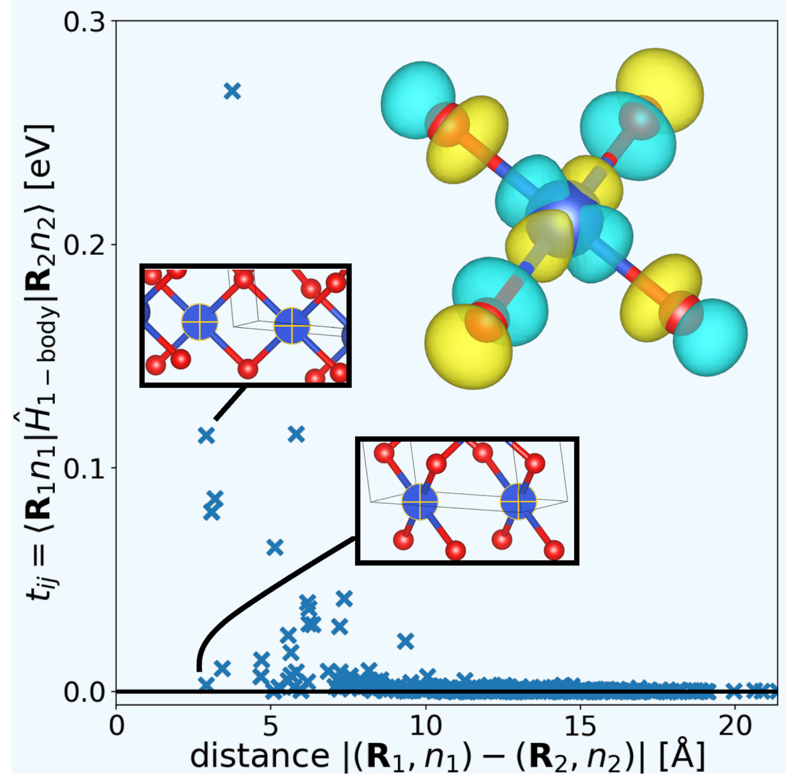
Near the Fermi level: **four conduction bands of Cu-3d$_{x^2-y^2}$ character** (from crystal field splitting), hybridized with O-2p orbitals.

Bands partially filled, yet CuO is insulating in experiments

$\Rightarrow$ **correlation effects**

Wannier Functions (WF)

WF are Fourier transform of the BS : $|\mathbf{R}n\rangle = \int_{BZ} d\mathbf{k}\, e^{-i\mathbf{k}\mathbf{R}} |\phi_{n,\mathbf{k}}\rangle$.

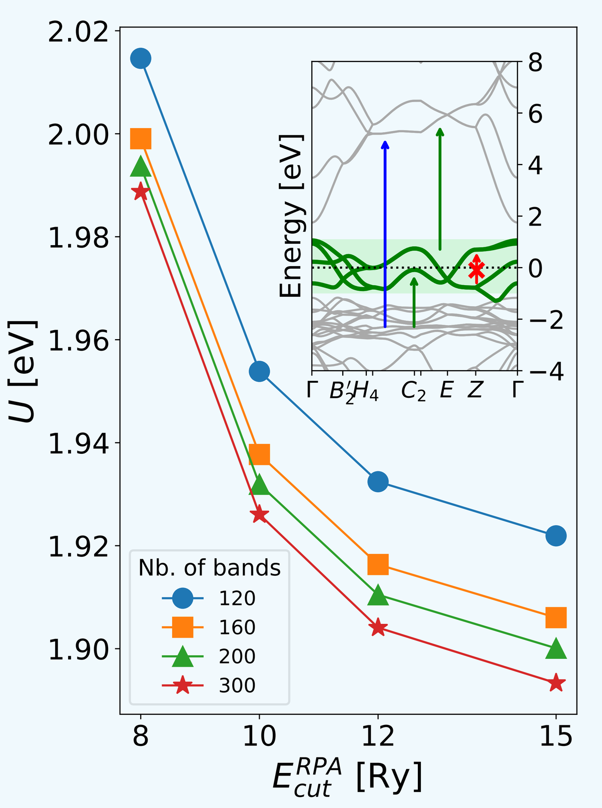


These WF are localized 3d$_{x^2-y^2}$ orbitals centered on each of the four copper atoms at lattice vector **R** (see top-right image).

Combination with the neighboring oxygen 2p orbitals, with σ-type overlaps

constrained Random Phase Approx.

Screening potential : $W(\mathbf{r}, \mathbf{r}') = (1 - vP_0^r)^{-1} v$



v : bare Coulomb interaction

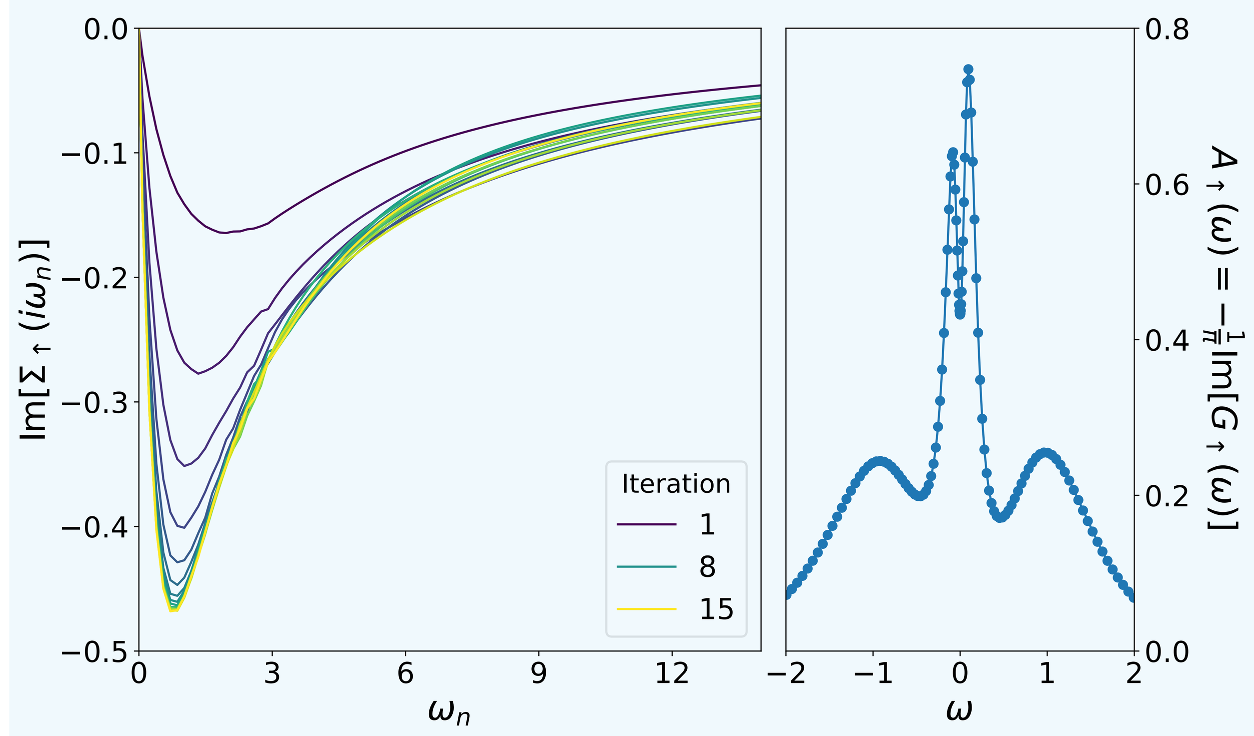
P_0^r : response function of W calculated using cRPA. Allowed transitions: outside the d-subspace (top right figure) and below E_{cut}^{RPA}.

Convergence graph

$U = 1.9$ eV for 200 bands and 15 Ry cutoff — value used in the following.

Dynamical Mean Field Theory (DMFT)

DMFT maps the lattice problem self-consistently onto an impurity model [3], solved with TRIQS & CTHYB [4], at $T = 290$ K in a paramagnetic phase (see right: imaginary part of the self-energy).



Analytic continuation from imaginary to real frequencies using MaxEnt [5] $\to$ spectral function $A(\omega)$ (right).

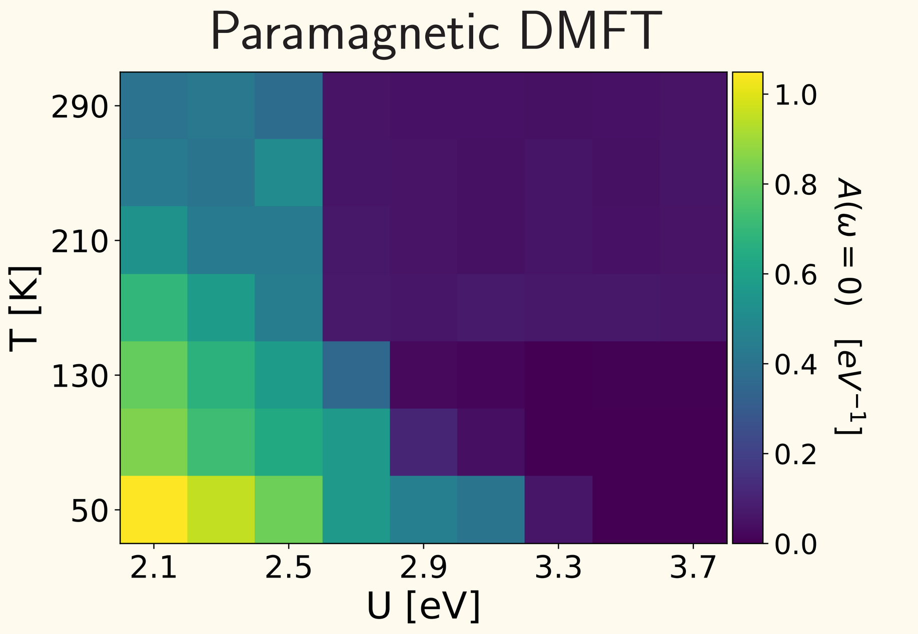
Non zero Spectral weight at $\omega = 0 \Rightarrow$ Bad metal phase.

Reason : overscreening of cRPA or missing antiferromagnetism ?

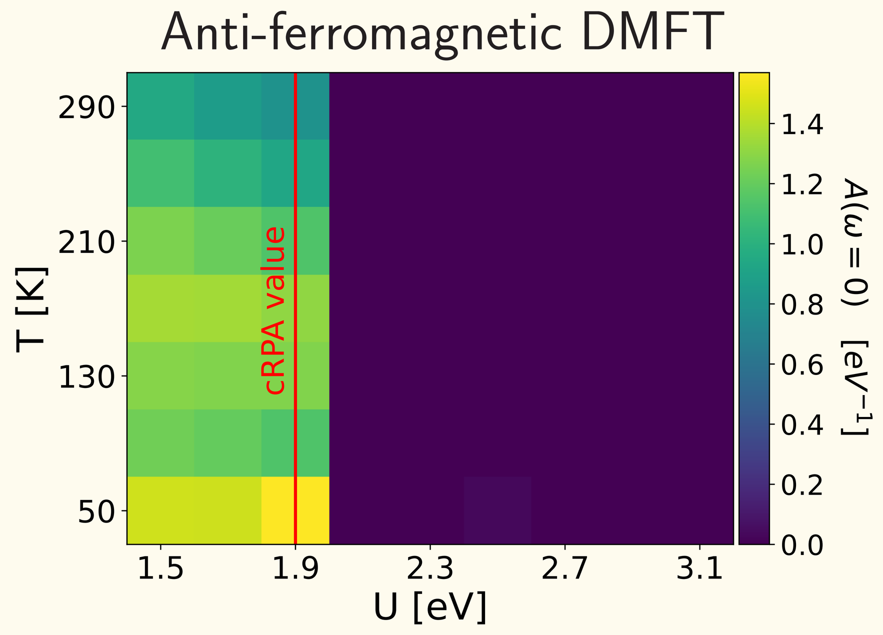
Phase Diagram of CuO

We computed the phase diagram of CuO in the paramagnetic and antiferromagnetic phase to choose the best parameters (U, T) that describe the insulating phase.

Paramagnetic DMFT



Anti-ferromagnetic DMFT



Ongoing developments

Last remaining step is the diagonalization of the impurity Hamiltonian, taking into account quadrupole $3s \to 3d$ excitations

$$\hat{H}_{imp} = \overbrace{\sum_{l\sigma} \epsilon_l a^\dagger_{l\sigma} a_{l\sigma}}^{\text{bath}} + \overbrace{\sum_{l\sigma} \left(V_{l\sigma} a^\dagger_{l\sigma} c_\sigma + \text{h.c.} \right)}^{\text{hybridization imp. - bath}} + \overbrace{E_{3s}\left(n_{3s\uparrow} + n_{3s\downarrow}\right)}^{\text{core}} + \underbrace{U n_\uparrow n_\downarrow - \mu\left(n_\uparrow + n_\downarrow\right)}_{\text{impurity}} + \underbrace{U_{3s,3d}\left(n_\uparrow n_{3s\uparrow} + n_\downarrow n_{3s\downarrow}\right)}_{\text{core-hole interaction}}$$

where $\{\epsilon_l, V_{l\sigma}, \mu\}$ are obtained from DMFT, and $\{U_{3s\to3d}, E_{3s}\}$ are taken from the literature. These eigenenergies and eigenstates can then be used to calculate the NRIXS cross section.

References

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[2] Right figure of the *Motivation* box : experimental results of orbital imaging of copper oxide at 300 K, by Victor Baledent and Jean-Pascal Rueff, with whom we are collaborating. The NRIXS spectra comes from paper [1] for NiO.

[3] A. Georges, G. Kotliar, W. Krauth, and M. J. Rozenberg. *Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. Rev. Mod. Phys.*, 68 :13–125, Jan 1996.

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